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BSH GUIDELINE

Early diagnosis of AL amyloidosis in haematology, cardiology, neurology, renal and general clinics: A British Society for Haematology Guideline

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Summary
AL amyloidosis is caused by a plasma cell clone that produces abnormal light chains that misfold and deposit as amyloid fibrils in tissues and thus affect organ function. The survival of patients with systemic amyloid light-chain (AL) amyloidosis largely depends on the extent of end-organ damage. Patients with early-stage disease can expect approximately 80% survival at 5 years with contemporary treatment, compared to less than 30% for those with advanced disease. The aim of this guideline is to facilitate recognition and raise suspicion of amyloidosis. It is important to recognise that amyloidosis can present in any clinic, including general medicine/care of the elderly and general practice.

KEYWORDS
amyloidosis, cardiac failure, MGUS, myeloma, proteinuria

METHODOLOGY

This guideline was prepared following the BSH process at (https://b-s-h.org.uk/media/16732/bsh-guidance-development-process-dec-5-18.pdf). The Grading of Recommendations Assessment, Development and Evaluation (GRADE) nomenclature was used to evaluate the levels of evidence and to assess the strength of recommendations. The GRADE criteria can be found at http://www.gradeworkinggroup.org.

LITERATURE REVIEW DETAILS

The literature search was undertaken with the help of BSH on 11 August 2023 using the PubMed database. A further literature search was undertaken by BSH on 28 February 2026 before submission. The keywords used were amyloidosis with one or more of the following: presymptomatic diagnosis, cardiac failure HFpEF, nephrotic syndrome, peripheral neuropathy, autonomic neuropathy, monoclonal gammopathy of clinical significance (MGCS),

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monoclonal gammopathy of renal significance (MGRS), monoclonal gammopathy of uncertain significance (MGUS), coagulopathy, macroglossia, soft tissue involvement and hepatomegaly. The search was further filtered using the following: inclusion criteria (human, clinical trial) and exclusion criteria (no papers published in non-English journals, no publications without an abstract).

REVIEW OF THE MANUSCRIPT

Review of the manuscript was performed by the British Society for Haematology (BSH) Guidelines Committee Haematology-Oncology, the BSH Guidelines Committee and the sounding board of BSH. It was also on the members section of the BSH website for comment. It has also been reviewed by UKMS and Myeloma UK; these organisations do not necessarily approve or endorse the contents.

INTRODUCTION

Amyloidoses are conformational diseases caused by misfolding and aggregation of proteins that deposit in tissues in the form of amyloid fibrils.[1,2] Incidence rates for all types of amyloidosis have been increasing due to better diagnostic techniques such as cardiac imaging, especially in the elderly for cardiac transthyretin amyloidosis (ATTR) amyloidosis, which is now the commonest acquired type of amyloid disease.

Amyloid light-chain (AL) amyloidosis is caused by a plasma cell clone that produces abnormal light chains that misfold and deposit as amyloid fibrils in tissues and thus affect organ function.[3,4] The majority of patients with AL amyloidosis have plasma cell infiltration of the bone marrow of less than 10% (i.e. MGUS like phenotype).[4] These low-grade, otherwise benign, clones that cause life-threatening end organ damage have been termed 'small dangerous plasma cell clones'.[5]

The survival of patients with systemic AL amyloidosis largely depends on the extent of end organ damage. Patients with early stage disease can expect approximately 80% survival at 5 years with contemporary treatment, compared to less than 30% for those with advanced disease.[6,7] Cardiac involvement remains the main cause of mortality but renal amyloidosis nonetheless causes significant morbidity due to progressive renal failure; dialysis is required in 75% of patients with stage III renal disease at presentation.[8,9] (Table 3). AL amyloidosis is often diagnosed late: in a patient experience survey, the diagnosis was made at least a year after the onset of symptoms in nearly 40% of the patients.[10]

At the UK National Amyloidosis Centre, 50.4% of all newly diagnosed patients with AL amyloidosis have stage III disease and 15% have very advanced stage IIIb disease.[11] An updated staging system (AL-ISS, AL amyloidosis international staging system) has recently been adopted as the new standard adding global longitudinal strain (GLS) to the previously defined stage IIIB, now termed as stage IIIc with a median survival of approximate 4 months.[12] These patients presenting with advanced irreversible organ involvement have a very high risk of death in the first few months despite modern treatments due to established advanced cardiac damage.[2]

The availability of highly sensitive biomarkers of clonal and organ involvement as well as better imaging have allowed for accurate risk stratification of the disease and is reshaping the approach to patients with AL amyloidosis.[4,13–14] We would like to emphasise that these biomarkers can also be used as tools for early identification of the disease.

The clinical manifestations of systemic AL amyloidosis are protean and a consequence of advanced organ damage. Most organs can be involved by AL amyloidosis (Figure 1) and a key diagnostic clue is the involvement of two or more systems. The clinical presentation and outcomes depend upon the pattern and severity of organ involvement by the amyloidogenic monoclonal protein, that is, light chain.[14] Diagnosis can be difficult as symptoms can mimic more common conditions.[15]

The aim of this guideline is to facilitate recognition or raise suspicion of amyloidosis. It is important to recognise that amyloidosis can present in any clinic, including general medicine/care of the elderly and general practice. The primary intended audience includes primary and secondary care non-haematology physicians with haematologists as secondary users.

The screening investigations for amyloidosis are listed in Table 1. Early diagnosis is paramount in improving outcomes as shown in Tables 2 and 3. Cardiac and renal staging at diagnosis in Table 2[12] and Table 3[8,9] are primarily prognostic and shown here to emphasise the need for early diagnosis as prognosis depends heavily on the stage at diagnosis.

A recent analysis of 3349 patients with AL from the ALChemy study shows that two-thirds of all renal AL patients still present with stage II or III disease, while the proportion of cardiac patients with stage IIIa disease has improved with increased awareness of amyloidosis in the cardiac community (Wechalekar, personal communication, NAC unpublished data 2026).

WHEN TO SUSPECT AL AMYLOIDOSIS IN PATIENTS WITH A PARAPROTEIN OR ABNORMAL FREE LIGHT CHAINS

AL amyloidosis is commoner than often appreciated and is a well-established complication of monoclonal gammopathies. Patients with lambda light chain disease are at particular risk, but AL amyloidosis can occur in those with kappa light chain disease or an intact paraprotein without light chain excess.[2]

Biomarker-based screening can help identify presymptomatic patients with monoclonal gammopathies who are developing AL amyloidosis.[4,17] AL amyloidosis was confirmed in a presymptomatic stage in 36% (22/60) of patients with MGUS with altered serum free light ratio (sFLR) who

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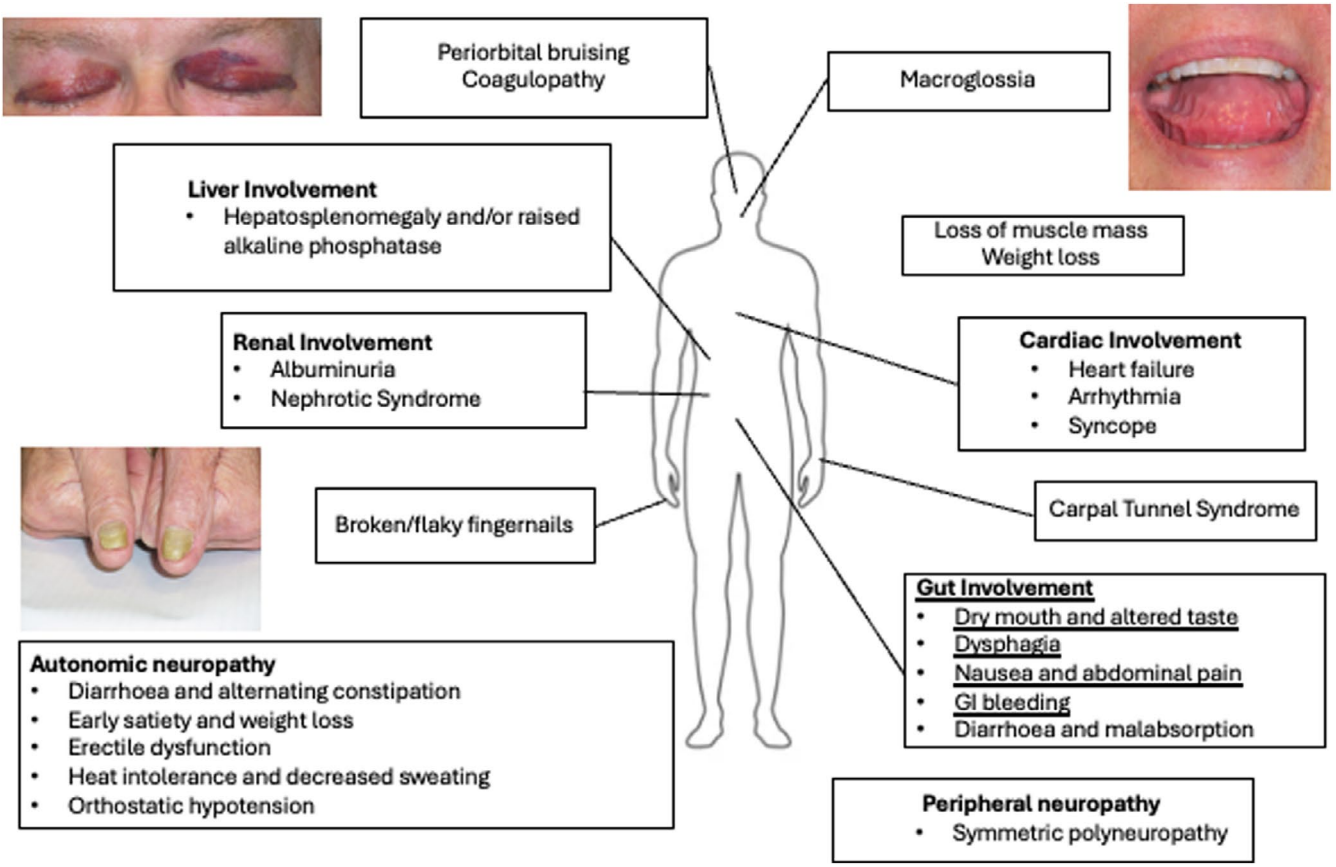


FIGURE 1 Clinical presentations of AL amyloidosis.

TABLE 1 Screening investigations for AL amyloidosis in non-haematology clinics.

- Immunoglobulin levels and serum protein electrophoresis (SPE) and immunofixation (IFE)
- Serum free light chain (FLC) assay to measure serum kappa and lambda levels and calculate the FLC ratio
- Urine protein electrophoresis and immunofixation looking for urinary Bence Jones protein (BJP)
- Urine albumin:creatinine ratio (ACR) and protein:creatinine ratio (PCR)
- U&E (urea and electrolytes)
- LFTs (liver function tests) serum albumin, bilirubin, alanine transaminase (ALT) and alkaline phosphatase (ALP)
- Bone profile: Serum calcium, albumin, phosphate and ALP
- Coagulation screen (prothrombin time [PT] and activated partial thromboplastin time [APTT])
- NT-proBNP (N-terminal pro-B-type natriuretic peptide) or BNP
- Troponin (Troponin T [TnT] or Troponin I [TnI])[a]
- ECG (electrocardiogram)

[a]The scoring systems for AL predominantly used TnT; however, many centres will now use alternative assays, for example, TnI and we recommend using whichever test is available locally. For the purposes of screening, elevation above the reference range of your local test (corrected for renal function) should raise concern. Care should be taken if TnT is requested outside of a specialist clinic setting as a raised result may prompt concern about acute coronary syndromes. The patient should also be informed that this test is being used as a staging blood test for a different condition.

TABLE 2 Staging of cardiac amyloidosis.[12,16]

	Staging of cardiac amyloidosis	Median survival
Stage I	NT-proBNP <332 ng/L and troponin T <0.035 ng/L	Not reached at 5 years
Stage II	NT-proBNP >332 ng/L but <8500 ng/L Or troponin T ≥0.035 ng/L	Not reached at 5 years
Stage III A	NT-proBNP <8500 ng/L and troponin T ≥0.035 ng/L	76 months
Stage III B	NT-proBNP >8500 ng/L and troponin T ≥0.035 ng/L	26 months
Stage III C[12]	NT-proBNP >8500 ng/L, troponin T ≥0.035 ng/L and global longitudinal strain >−9%	4 months

Abbreviation: NT-proBNP, N-terminal pro-B-type natriuretic peptide.

had a biomarker abnormality in their follow-up for MGUS in a large single-centre study of 4742 patients with MGUS with a median follow-up of 4 years.[17] Notably, in this study, none of the MGUS cases without a biomarker alteration progressed to AL amyloidosis.

Initial investigations for patients referred with a paraproteinaemia should also include urine protein:creatinine ratio and albumin:creatinine ratio (PCR and ACR), N-terminal pro-B-type natriuretic peptide (NT-proBNP), a troponin assay and alkaline phosphatase (ALP) to screen for early renal, cardiac and liver involvement respectively.

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TABLE 3 Staging of renal amyloidosis.[8,9]

	Staging of renal amyloidosis	Risk of going on dialysis/renal survival
Stage 1	Proteinuria ≤5 g/24 h and eGFR ≥50	None
Stage 2	Either proteinuria >5 g/24 h or eGFR <50	11%–25% at 2 years
Stage 3	Both proteinuria >5 g/24 h and eGFR <50	60%–75% at 2 years

Abbreviation: eGFR, estimated glomerular filtration rate.

One should suspect kidney involvement if 24 h proteinuria >0.5 g/24 h or urine PCR >50 mg/mmol with a concordant rise in ACR >30 mg/mmol, suspect cardiac involvement if NT-proBNP >332 ng/L, suspect liver involvement if alkaline phosphatase >1.5 times the institutional upper limit of normal.

A detailed clinical history to identify other organ involvement should be part of the assessment.

NT-proBNP is always raised in cardiac AL amyloidosis and high levels can precede the development of clinically evident cardiac AL amyloidosis by many years. In an MGUS/myeloma clinic, unexplained high NT-proBNP >332 ng/L, or progressively rising levels, should raise a high suspicion of cardiac amyloidosis.[16–18] NT-proBNP may be falsely raised in atrial fibrillation, renal impairment and elderly patient.

The first manifestation of renal amyloidosis is frequently albuminuria. Urine PCR and ACR have superseded 24-h urine collection as easier tests in practice. Twenty-four hours of urine protein is cumbersome to perform, inconvenient for patients and subject to sample collection variability. Urine PCR is non-specific and will detect all proteinuria (including albumin as well as monoclonal light chains i.e. Bence Jones protein). In contrast, ACR specifically detects albuminuria and is a better test for assessing glomerular disease.[17,19–20] Unexplained urine PCR of 50 mg/mmol that is roughly equivalent to a urinary protein excretion of 0.5 g/24 h should raise a suspicion of renal amyloidosis if urine ACR is also raised and >30 mg/mmol. Discordant rise in urine PCR without a rise in ACR in an MGUS patient can be due to urine light chains also called Bence Jones proteinuria.

ALP was found to be raised in 86% of patients with hepatic amyloidosis. Weight loss and hepatomegaly are well-known late features of hepatic amyloidosis and raised bilirubin is a poor prognostic marker.[21]

Healthcare professionals who monitor patients with MGUS or smouldering myeloma for progression should be aware that patients can progress to AL amyloidosis with minimal or no increase in clonal light chains and clinical vigilance is of the utmost importance.

Therefore, in addition to monitoring for more usual criteria for progression to myeloma such as bone pain, anaemia and renal impairment, one must be watchful for unexpected symptoms, signs or investigation findings that could be consistent with AL amyloidosis as described in Figure 1. For example, new symptoms such as pedal oedema, unexplained severe fatigue, weight loss, dry mouth, difficulty in chewing or swallowing, postural hypotension, frothy urine and shortness of breath are of equal importance to the classical features of myeloma.

We advise adding urine ACR and PCR and NT-proBNP to standard tests (serum protein electrophoresis (SPE), immunofixation (IFE) and serum free light chains (SFLC), full blood count (FBC), urea and electrolytes (U&E), bone profile and liver function tests (LFT)), at baseline assessment as well as at every monitoring assessment of a paraprotein as recommended by the recent MGUS and smouldering myeloma good practice papers. A troponin assay should be considered if the NT-proBNP is abnormal.[22,23]

If AL amyloidosis is suspected, a histological diagnosis should be sought and appropriate amyloid typing undertaken. Congo red positivity in biopsy from any site (including bone marrow) should prompt detailed assessment for end organ involvement.

Although beyond the specific remit of this guideline, other M-protein-associated disorders should also be borne in mind such as monoclonal gammopathy of renal significance (MGRS), light chain deposition disease (LCDD) and POEMS syndrome (polyneuropathy, oedema/organomegaly, endocrinopathy, monoclonal gammopathy, skin changes).

Recommendation for patients in Monitoring Clinics for MGUS:

1. **Urine PCR and ACR and NT-ProBNP should be added to standard tests for assessment as well as monitoring of MGUS (Grade 1A).**
2. **Falling serum albumin levels and raised alkaline phosphatase should be noted and if persistent, then should be investigated further (Grade 1A).**
3. **Patients can progress to AL amyloidosis with minimal or no increase in clonal light chains (Grade 1A).**

WHEN TO SUSPECT AL AMYLOIDOSIS IN PATIENTS WITH UNEXPLAINED CARDIAC IMPAIRMENT

The key features that should raise a suspicion for amyloidosis in patients with heart failure with or without a known monoclonal gammopathy:

- Low serum albumin (<30 g/L);
- Unexplained persistent proteinuria (urine PCR or ACR >50 mg/mmol);
- Autonomic and/or peripheral neuropathy;
- Unexplained persistently high (or rising) troponin without significant coronary artery disease;
- Significant persistent hypotension or unexplained worsening of heart failure symptoms after starting angiotensin-converting enzyme inhibitors (ACE-I) or angiotensin receptor blockers (ARBs), beta-blockers or angiotensin receptor-neprilysin inhibitor (ARNi);

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- Previously hypertensive patient developing unexplained low or low normal blood pressure despite reduction of anti-hypertensives;
- Heart failure with preserved ejection fraction (HFpEF) on echocardiogram (ECHO);
- Low-voltage complexes on electrocardiogram;
- History of bilateral carpal tunnel surgery; biceps (or other) tendon rupture; spinal canal stenosis;
- Unexplained weight and muscle loss (cardiac cachexia);
- Echocardiographic features:
    - 'Left ventricular hypertrophy' with normal or low voltage ECG;
    - Loss of longitudinal function, that is, abnormal global longitudinal strain (GLS) with apex to base gradient with a preserved apical compared to basal function.

A high index of suspicion should be maintained while interpreting echocardiograms and cardiac magnetic resonance imaging (MRI). Radiographic changes may be subtle, and it is important that imaging is undertaken by a practitioner with expertise in imaging of cardiac amyloidosis. An alternative aetiology for cardiac amyloidosis is ATTR. It is important to remember that ATTR and MGUS may co-exist.[24] There are guidelines for differentiation and evaluation of types of cardiac amyloidosis.[25] Screening investigations for a monoclonal gammopathy should be undertaken in all patients with suspected cardiac amyloidosis, as listed in Table 1.

Recommendation for patients presenting with cardiac failure in cardiology clinics:

1. **Amyloid should be considered as a differential cause of unexplained cardiac failure (Grade 1B).**
2. **Screening investigations for AL amyloidosis should be undertaken in all patients with unexplained cardiac failure, especially with preserved ejection fraction (Grade 1B).**
3. **Every effort should be made to clarify the type of amyloid deposits in any patient with cardiac amyloidosis in a patient with MGUS at a specialist centre as MGUS and unrelated ATTR frequently coexist. Please note that both AL and ATTR may also occur concurrently (Grade 1A).**

WHEN TO SUSPECT AL AMYLOIDOSIS IN PATIENTS WITH RENAL IMPAIRMENT

Testing for myeloma and screening for AL amyloidosis (see box 1) should be undertaken in all of the following groups:

- Unexplained persistent proteinuria;
- Urine PCR/ACR >50 mg/mmol;
- Progressive renal impairment after the age of 40 (https://bestpractice.bmj.com/topics/en-gb/875);
- High clinical suspicion of AL amyloidosis.

Up to 50% of patients with a monoclonal gammopathy with proteinuria >1.5 g/L or microscopic haematuria (for non-AL MGRS), particularly associated with an abnormal FLC ratio, will have a monoclonal protein-related renal disorder. Unless there is a contraindication or a high suspicion of AL amyloidosis, all patients should be referred for a kidney biopsy. In suspected non-AL MGRS, a kidney biopsy is more informative than a bone marrow biopsy, whereas in AL, a bone marrow test or fat pad aspirate or biopsy is less invasive and can be undertaken before a kidney biopsy. A renal biopsy is not routinely indicated in patients with monoclonal gammopathy and a significant proteinuria if the bone marrow or fat pad biopsy shows Congo red positivity and the amyloid type is confirmed to be AL type as the kidney involvement is most likely to be due to renal amyloidosis.

Unexpected features like blood in the urine or non-proteinuric chronic kidney disease (CKD) need further investigations.

A known monoclonal gammopathy should lower the threshold for screening for myeloma, AL amyloidosis and MGRS. However, in a large published series of patients with unexplained proteinuria undergoing a kidney biopsy who subsequently were diagnosed with MGRS, 29% of patients were not known to have a monoclonal paraprotein prior to the biopsy and >70% of patients with a subsequent diagnosis of amyloidosis were not known to have a monoclonal protein prior to biopsy.[26]

Recommendation for patients presenting with a paraprotein in renal clinics:

1. **Urine PCR and ACR can be a surrogate for 24-h urine collection for quantitation of proteinuria and albuminuria; urine ACR and urine PCR should be monitored in all cases of MGUS[19,20] (Grade 1A).**
2. **Kidney biopsy is the most definitive test to make a diagnosis of MGRS and AL amyloidosis affecting kidneys and should be considered in any patient with MGUS/SMM with unexplained rise in urine ACR > 50 (Grade 1A).**
3. **A kidney biopsy can be avoided in the presence of MGUS/SMM and confirmation of AL amyloid deposits on another biopsy (bone marrow or fat pad biopsy) (Grade 2B).**

WHEN TO SUSPECT AL AMYLOIDOSIS IN UNEXPLAINED NEUROPATHY

Both AL amyloidosis and ATTR can involve the peripheral nervous system. In addition, monoclonal gammopathies, particularly immunoglobulin M (IgM) gammopathies, may result in paraproteins with direct anti-neuronal antibody activity.[27] We, therefore, recommend that assessment for a monoclonal protein (see Table 1) forms part of the routine diagnostic work-up for patients with unexplained

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polyneuropathy. A clinical haematologist should be involved if AL amyloidosis is suspected.

A combination of peripheral and autonomic neuropathy is a hallmark of systemic amyloidosis (AL or ATTR). The presence of both without an apparent cause (e.g. long-standing uncontrolled diabetes) should prompt a detailed search for amyloidosis. Conversely, isolated non-progressive or very slowly progressive neuropathy with a monoclonal protein but normal serum free light chains is rarely caused by amyloid deposition.

A careful history should be conducted to look for autonomic features including diarrhoea, erectile dysfunction and orthostatic intolerance that may be exacerbated by large meals or alcohol, or the introduction of anti-hypertensive or other cardiac medications, or alpha-blockers for prostatic hypertrophy. Prominent or nocturnal diarrhoea is a red flag for an amyloid neuropathy.[28]

Recommendations for a patient presenting with a paraprotein in neurology clinics:

1. **The combination of peripheral and autonomic neuropathy without a cause should prompt screening for amyloidosis (Grade 1A).**
2. **A sustained fall in blood pressures by 20/10 mm Hg by 3 min with little compensatory rise in heart rate suggests neurogenic orthostatic hypotension (Grade 1A).**
3. **Concurrent carpal tunnel syndrome in addition to other systemic features of AL amyloidosis as shown in Figure 1 should raise the clinical suspicion for AL (Grade 1A).**

WHEN TO REFER URGENTLY TO A HAEMATOLOGY CLINIC

Consideration should be given to referring patients urgently for haematology review with a confirmed (or highly likely) diagnosis of AL amyloidosis with the following clinical manifestations:

- Hypoalbuminaemia <25 g/L;
- Falling estimated glomerular filtration rate (eGFR) <50 mL/min/1.73 m^2;
- All cases of cardiac amyloidosis;
- Autonomic neuropathy with postural hypotension >20 mmHg;
- Hepatic amyloidosis with high bilirubin (>2 × upper limit of normal)[21];
- Bleeding or bruising due to Factor X deficiency.

SUMMARY

AL amyloidosis is a chameleon of clinical medicine; its aetiology is complex, manifestations myriad and management subspecialised. It is not, however, particularly rare. It is therefore important that clinicians who may encounter patients with this disorder are aware of the disease and able to detect it. We have here summarised key features of the disease and outlined suggested investigations that will aid diagnosis and enable early referral which improves outcomes for patients with this condition.

Once diagnosed, all patients with AL amyloidosis should be discussed in a multidisciplinary meeting experienced in diagnosing and treating AL amyloidosis. All patients with AL amyloidosis should be referred to the National Amyloid Centre for confirmation of diagnosis, treatment advice and encouraged to take part in clinical trials and contribute to centralised data collection.

AUTHOR CONTRIBUTIONS

All authors contributed to the conceptualisation, methodology and writing and review of the guidelines.

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CONFLICT OF INTEREST STATEMENT

All authors have declared a conflict of interest disclosed to the BSH and Task Force Chairs, which may be viewed on request. MG: advisory board, grant for service improvement, speaker honoraria and conference support from Alnylam, Amgen, Celgene BMS, Janssen, Novartis Pharma&, Pfizer, Sanofi, Sobi, Stemline therapeutics and Takeda. SC and JR: no conflicts of interest. CP: advisory board, speaker honoraria and conference support from Abbvie, Amgen, BMS, Everything Genetic, Gilead. Janssen, Jazz, Novartis, Pfizer, Sanofi. SH: speaker honoraria and conference support from Amgen, Celgene, Takeda and Janssen. HL: advisory board, speaker honoraria and conference support from Alnylam, Alexion Janssen, Novartis, Pfizer and SOBI. LC: speaker honoraria, conference support and consultancy fees from Alnylam, AstraZeneca, Boehringer Ingelheim, Novartis, Pfizer, Pharmacosmos, Vifor. TC: consultancy fees from Aptitude Health. AW: speaker honoraria, conference support and consultancy fees from Attralus, Alexion, AstraZeneca, GSK, Janssen, Pfizer and Prothena.

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DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no datasets were generated or analysed for this guideline.

REVIEW PROCESS

Members of the writing group will inform the writing group chair if any new evidence becomes available that would alter the strength of the recommendations made in this document or render it obsolete. The document will be reviewed regularly by the relevant task force and the literature search will be re-run every 5 years to search systematically for any new evidence that may have been missed. The document will be archived and removed from the BSH current guidelines website if it becomes obsolete. Please check the BSH guidelines website for any addenda that may be produced after the initial publication.

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AUDIT TOOL

Blank audit template can be found for the writing group to complete: https://b-s-h.org.uk/media/15658/audit-template-mar-2017.doc.

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